

Enhanced Network Security Protocols for the Quantum Era: Combining Classical and Post-Quantum Cryptography, and Quantum Key Distribution

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Abstract—The emergence of quantum computing poses a threat to classical cryptography algorithms, necessitating a shift to quantum secure cryptography. Hybrid protocols combining at least one classical and one quantum-resistant cryptographic algorithm are becoming the standard for securing communications. In this work, we present our novel solution for integrating three different cryptographic assumptions (two of them quantum-resistant) into hybrid network security protocols, ensuring that three different cryptographic assumptions must be broken before the protocol becomes vulnerable. Our solution allows for a seamless integration of classical and post-quantum (PQ) cryptography, and quantum key distribution (QKD) into existing network security protocols (e.g., TLS, IPsec) without any major modifications to the protocols themselves. This crypto-agility ensures the mitigation of some of the most well known challenges of both PQ cryptography and QKD. Our findings demonstrate the feasibility of such triple-hybrid network security protocols, showing non-substantial decrease in performance and almost no added packet overhead compared to state of the art protocols. In exchange, we pave the way towards next generation networks where the potential of new quantum-resistant cryptographic schemes can be leveraged in a dynamic and agile fashion, thus fostering a new era of unbreakable communication systems.

Index Terms—Quantum-resistant cryptography, quantum key distribution, post-quantum cryptography, classical cryptography, triple-hybrid authenticated key exchange.

I. INTRODUCTION

QUANTUM-RESISTANT communications have emerged as a response to the advancements of quantum computers. Shor's algorithm has demonstrated that, given a powerful

enough quantum computer, problems such as factorization and discrete logarithms could be solved exponentially faster compared to the best known classical algorithms (e.g., GNFS) [1]. This translates into a need to substitute currently used public key cryptography (PKC) algorithms. Specifically, Rivest-Shamir-Adleman (RSA) and elliptic curve (EC) cryptography algorithms must be substituted by new, quantum-resistant alternatives [2], [3], [4].

Although the timeline for such a powerful quantum computer to be available remains uncertain, the migration of communications systems towards implementing quantum-resistant schemes should be performed as soon as possible, as it presents several challenges, some of which we address in this work, that require a timely controlled transition [5].

Moreover, “harvest now, decrypt later” (HNDL) attacks allow adversaries to already collect encrypted versions of long-lived data today, which can later on be decrypted by means of a quantum computer. Thus, not only will classical cryptographic schemes be compromised in the near future, but the confidentiality of current communications is already at risk.

In anticipation of future cryptographic challenges, standards development organizations (SDOs) such as NIST, ETSI, ITU-T, ISO/IEC, IETF, and IEEE are actively working on evolving to the next era of quantum-resistant cryptographic schemes. Among the most developed alternatives to replace traditional PKC algorithms are post-quantum (PQ) cryptography and quantum key distribution (QKD).

While PQ cryptography relies on mathematical assumptions (e.g., lattice problems, hash functions) for which Shor's algorithm provides no known advantage, QKD relies on the delivery of quantum signals among two authorized partners; in this scenario, eavesdropping on the key distribution can be detected, and thus the key distribution protocol can be aborted. Otherwise, if no eavesdropper is detected, the shared secret transmitted between Alice and Bob can be considered information-theoretically secure (ITS) [6].

The two approaches fundamentally differ in their security paradigms: PQ cryptography maintains computational security against quantum computers via different mathematical assumptions, while QKD aims for ITS security during key distribution. Despite their different mechanisms, both strate-

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gies effectively mitigate the advantages that Shor's algorithm provides against classical PKC.

Nevertheless, it is worth mentioning that the security of subsequent use of these keys for symmetric encryption depends on the method used. Computationally secure symmetric encryption, such as the Advanced Encryption Standard (AES) algorithm, remains 'only' computationally secure, but is still considered secure against quantum computers if sufficiently large keys are employed. To achieve ITS secure communications, QKD must be combined with an ITS encryption primitive, such as one-time pad (OTP). However, the large key sizes required by OTP – at least as long as the message itself – limit its practical use. Given the existing challenges of both PQ cryptography and QKD, combining PQ cryptography and QKD can create a synergistic approach, leveraging the strengths of both technologies and providing a more robust security against quantum threats.

Our work presents a novel quantum-resistant software solution that allows the seamless integration of classical and PQ cryptography, and QKD to generate a single, shared, secret key to be used for symmetric encryption. By combining the shared secrets generated by the aforementioned cryptography techniques, we can achieve a crypto-agile, resilient protocol that leverages the benefits of three independent cryptographic assumptions, immediately addressing HNDL attacks, while paving the road for hybrid and triple-hybrid quantum-resistant protocols to be implemented into next generation networks.

The proposed solution enhances the security level of both Transport Layer Security (TLS) and Internet protocol security (IPsec), which represent the security backbone of many modern networks, including public and corporate virtual private networks (VPNs) and fifth generation (5G) mobile communications [7], [8]. Our solution requires three different cryptographic assumptions (two of them quantum-resistant) to be broken before the system becomes vulnerable to attacks. The main contributions of our work are as follows:

- To the best of the authors' knowledge, we are the first to provide a network security solution able to combine classical and PQ cryptography, and QKD and seamlessly integrate such hybrid cryptographic techniques into multiple, well known network security protocols without major modifications to the protocols themselves.
- We demonstrate novel implementations of state of the art, open source IPsec and TLS 1.3 protocols (e.g., strongSwan [9], OpenSSL [10]), in which quantum resilience is achieved by combining classical cryptography, PQ cryptography, and QKD.
- Our work provides ready-to-use quantum-resistant TLS 1.3 and IPsec solutions that can already be employed to prevent one of the most urgent threats posed by quantum computing – HNDL – across the security backbone of various network architectures, regardless of whether they are IPsec-based or TLS-based.
- Finally, the versatility of our solution can mitigate some of the most commonly-known challenges of both PQ cryptography and QKD, while paving the way for the standardization of hybrid, or even triple-hybrid network

security protocols based on classical, PQ cryptography and QKD.

The remainder of this manuscript is structured as follows: Section II provides an overview of the security requirements of next generation networks in the advent of quantum computers and proposes the use of hybrid protocols for enhanced protection. Section III discusses the transition from classical to hybrid quantum-resistant network security protocols. Section IV describes our approach to implement quantum-resistant TLS. In Section V, our approach for integrating our triple-hybrid solution into IPsec is explained. Section VI presents our results and provides a performance analysis for each implementation. Finally, Section VII summarizes and concludes the article.

II. FROM CLASSICAL TO QUANTUM-RESISTANT COMMUNICATIONS IN NEXT GENERATION NETWORKS

Network security is an increasingly critical concern in the digital age. Since the emergence of the Internet and mobile communications, and now with the latest advancements in artificial intelligence (AI) and large language models (LLM), a growing amount of confidential information is exchanged through channels secured by classical cryptography. However, the advent of quantum computing challenges these security assumptions, initiating a transition period where all encrypted communication systems must become quantum-resistant to effectively protect data in transit [11].

Current communications are secured via a mix of asymmetric PKC algorithms and symmetric encryption schemes. The former are used to negotiate a shared secret key, while the latter use the previously agreed shared secret key to implement symmetric encryption among two communicating parties. However, Shor's algorithm has demonstrated that, given a powerful enough quantum computer, problems such as factorization and discrete logarithms could be solved in polynomial time compared to sub-exponential time provided by the best known classical algorithms (e.g., GNFS) [1].

This speedup makes classical cryptography vulnerable and therefore, it must be replaced as soon as possible. While Shor's algorithm primarily challenges PKC cryptographic operations based on integer factorization and discrete logarithm problems, another class of quantum algorithms known as quantum search algorithms poses a potential threat to symmetric encryption [1], [12].

Quantum search algorithms are designed to find specific elements in unstructured databases more efficiently than classical algorithms, which could be applied to brute-force attacks on encryption keys. Among quantum search algorithms, Grover's algorithm is the most well known and studied [12]. It demonstrates a quadratic speedup over classical search algorithms, potentially impacting the security of symmetric encryption schemes like the AES algorithm.

Theoretically, Grover's algorithm could reduce the effective security of an n bit key to approximately $n/2$ bit. This means that a quantum algorithm can execute a brute-force attack on a block cipher with n bit keys using $2^{n/2}$ queries instead of the typical 2^n . For instance, it could reduce AES-128's effective

security to about 64 bits, and AES-256's to about 128 bits [13]. To mitigate the advancements provided by quantum computers – only considering Grover's algorithm – with regards to symmetric cryptography, it is therefore considered sufficient to employ AES-256 Galois-counter mode (AES-256-GCM) and to ensure that quantum-safe authenticated encryption is used.

While both PQ cryptography and QKD can protect a communication system against the attacks of a quantum computer during the key negotiation phase – when considering existing quantum algorithms – there are some challenges that need to be addressed before such systems can be employed in real life applications.

On the one hand, PQ cryptography raises concerns with regard to its maturity and the risk of future cryptanalytic attacks that might break PQ algorithms thought to be secure. As an example, Beullens [14] successfully demonstrated that the Rainbow algorithm, one of the third round finalist of the NIST PQ standardization process, and considered to be secure, was subject to a key recovery attack “feasible for an attacker with a moderate amount of resources” and that could be executed on an average of 53 hours [14], [15]. The Supersingular Isogeny Diffie-Hellman protocol (SIDH) has also been demonstrated vulnerable to key recovery attacks [16]. Another challenge of PQ cryptography is ensuring accurate security analyses. Furthermore, Bernstein [17] have recently filed a lawsuit questioning the standardization process of CRYSTALS-Kyber, after it was discovered that the security analysis of the Kyber-512 parameter set, currently known as ML-KEM-512, significantly underestimated the algorithm's security level [18], [19]. Alagic et al. claimed that such an oversight raises concerns about the entire NIST standardization process [15]. The resulting uncertainty, along with other issues intrinsic to PQ cryptography, such as the substantially larger key exchange payloads [4], are delaying the adoption of PQ cryptography at a global scale [20].

On the other hand, regarding QKD, the main challenge lies in the existing trade-off between secret key generation rate (SKR) and transmission distance between QKD nodes. As the distance between nodes grows, the corresponding loss of the optical fiber also increases (due to the physical properties of the optical fiber). This causes the SKR to decays exponentially, limiting the scalability of QKD networks on a global scale and necessitating the introduction of trusted nodes for relaying key material [21], [22].

Moreover, the seamless integration and coexistence of QKD networks into currently existing classical infrastructure is a significant challenge being addressed, as the interaction among quantum and classical signals over the same fiber has a dramatic effect on the SKR generation. In this regard, the coexistence of both technologies is crucial, as the cost of deploying dedicated dark-fiber networks solely for QKD is prohibitively costly for most network operators. To address this challenge, significant progress is being made towards overcoming the SKR-distance trade-off limitation of QKD, as well as exploring new methods to transmit classical and quantum signals over the same fiber with minimum added cost and impact on the performance of the SKR [23], [24], [25], [26].

Because of the aforementioned reasons, we understand that triple-hybrid quantum-resistant network security protocols based on classical, PQ cryptography and QKD will add an additional layer of defense against quantum computers, providing quantum-resilient defense-in-depth and allowing greater crypto-agility via various quantum-resistant cryptographic assumptions.

In our work, we study, implement and analyze such solutions in realistic network scenarios using state of the art equipment and open source libraries. Such an implementation can mitigate the potential vulnerabilities of PQ cryptography, allows a quick and agile replacement of any of the components involved in the case of a catastrophic failure, and leverages the potential of QKD primitives whenever possible, ensuring that several fallback mechanisms are available.

III. HYBRID SECURITY BASED ON CLASSICAL, PQ CRYPTOGRAPHY AND QKD

The following section explains the architectural requirements for developing hybrid solutions based on classical, PQ cryptography, and QKD. Due to the technological differences among the components involved in the key negotiation, we separately explain the hardware and software requirements for deploying such solutions, and provide an example of how this hybrid architecture fits into TLS and IPsec – the two network security protocols that protect the backbone of modern communication networks. Moreover, we discuss security proofs for triple-hybrid key combiners found in the literature, explaining the enhanced security assumptions of combining classical, PQ cryptography and QKD.

A. Infrastructure Requirements for Deploying Hybrid Security

Hybrid authenticated key exchange (AKE) protocols combine key material from different cryptographic sources to build network security protocols resilient to catastrophic failures, threatening technology advances, and future cryptanalytic attacks.

PQ cryptography and QKD are assumed to be secure against quantum computers, but they are still ‘relatively’ immature cryptographic technologies that must be subject to further deeper study for the following reasons: 1) it must be ensured that their security assumptions are correct; 2) the integration of such cryptographic techniques into current network security protocols must be seamless, i.e., with a minimum impact on performance; 3) protocols must be crypto-agile, meaning that in case of any future algorithmic discovery, or cryptanalytic attack that might affect the protocol's security, the underlying cryptographic assumptions can be replaced immediately.

By developing triple-hybrid security protocols, we can analyze the aforementioned requirements in a realistic network scenario and design network security protocols that not only meet these requirements but also enhance the overall security of communication systems. Classical and PQ cryptographic algorithms can potentially be deployed in almost any existing network equipment, as both solutions are completely software-based. To achieve maximum security in scenarios where

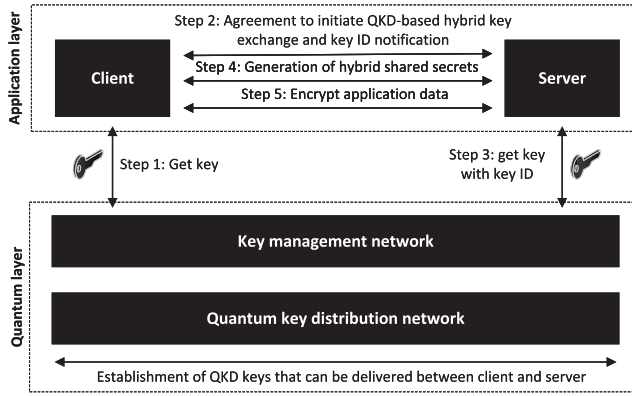


Fig. 1. Hybrid-security for client-server communications: Experimental architecture for combining QKD with classical and PQ cryptography, including the application, KMS and QKD layers [20].

security is the highest priority, QKD must also be employed, but taking into account its limitations.

QKD is a physical layer technology, and due to the physical constraints and challenges of current QKD technology (e.g., dedicated equipment, SKR-distance trade-off, cost of equipment, coexistence with classical network), certain elements must be available in order to successfully employ QKD. The infrastructure shown in Fig. 1 provides an overview of all functional elements that must be present in a communication system to be able to use classical, PQ and QKD keys for encrypting data.

First, both client and server must have access to the ‘quantum layer’ in a physically secure environment, so that the key retrieval can be physically secured [27]. Nowadays, the delivery of keys is performed via TLS, which might potentially use classical and PQ cryptography to perform the key retrieval. However, this would require the key delivery between the secure application entities (SAE) and the key management servers (KMS) to use a ‘weaker’ cryptographic algorithm, which would undermine the security of the key retrieval, and thus the whole QKD system.

Therefore, the current standard [27] establishes that the QKD node, KMS, and SAE must be in the same point-of-presence, ensuring that only authorized partners have physical access to the QKD equipment. This guarantees that, even though the key delivery uses classical or PQ cryptography, the assumption that the key delivery process is secure can be made.

Secondly, a functional key management network (KMN) layer must be operational, as it acts as a courier among the QKD nodes generating keys and the SAEs, ensuring that key delivery happens in a controlled and secure manner. In this regard, much work is currently being carried out on the topic of optimizing the KMN layer, so that QKD networks can evolve beyond point-to-point communications [28]. Finally, at the physical layer, QKD nodes must ensure the correct distribution of quantum signals among authorized partners, as described in [29].

To effectively implement the aforementioned hybrid QKD-based key exchange within network security protocols such

as TLS, IPsec, MACsec, or others, the protocols must ensure that at least five steps shown in Fig. 1 take place before QKD-enabled encrypted communications start between client and server:

- 1) **Get key:** Client must be able to communicate to the KMN layer and obtain both a QKD shared key and its corresponding key identifier (key ID)
- 2) **Agreement to initiate QKD-based hybrid key exchange and key ID notification:** Client and server must be able to agree on the use of QKD shared keys, and client must transmit the key ID to the server in order to retrieve the same shared secret.
- 3) **Get key with key ID:** With the key ID received by the client, the server must be able to obtain the same shared secret as the client.
- 4) **Generation of hybrid shared secrets:** After both client and server have negotiated the same QKD secret key, as well as classical and PQ secret keys, the three of them must be combined in a safe way in order to obtain the final encryption keys.
- 5) **Encrypt application data:** Once the final encryption keys have been successfully derived at both client and server endpoints, they can be employed via a quantum-resistant authenticated encryption with associated data (AEAD) algorithm to ensure quantum-resistant communications.

B. Security Assumptions of Triple-Hybrid Key Combiners

In the context of PQ cryptography, NIST has recently standardized a key encapsulation mechanism (KEM) algorithm called Module-Lattice-Based KEM (ML-KEM), formerly known as Crystals-Kyber, designed for quantum-resistant key exchange [18]. The security of ML-KEM is related to the computational difficulty of solving certain systems of noisy linear equations, specifically the module learning with errors problem (MLWE).

The hardness of the MLWE problem is based on the presumed hardness of specific computational problems in module lattices. By applying a Fujisaki-Okamoto (FO) transform, the MLWE problem can be converted into a KEM mechanism, satisfying the so-called indistinguishably under adaptive chosen-ciphertext attacks (IND-CCA) security property. As explained in [18], in order to inherit the IND-CCA security property, implementations should assess the security of combined KEMs that include ML-KEM so that hybrid exchanges still retain IND-CCA [30]. Three parameter sets (i.e., 512, 768, 1024) offer different trade-offs between security and efficiency. For instance, to break ML-KEM-1024 (e.g., 256 bits of security), a quantum computer – using Grover’s algorithm – would require 2^{128} operations [31], which is currently assumed to be quantum safe. A detailed analysis of the security assumptions of ML-KEM is shown in [18].

Regarding QKD, a well-established security model is presented in [32], confirming that QKD provides ITS security under rigorous theoretical conditions. Although QKD has been theoretically proven secure, practical implementations may be vulnerable to side-channel attacks [32]. While this remains

a potential issue, side-channel attacks are now routinely addressed in new QKD equipment, and secure implementations (e.g., DV-QKD, CV-QKD, MDI-QKD) are expected. Therefore, we can assume the shared secret generated via QKD to be ITS secure.

In the context of key exchange, PQ cryptography is explicitly designed to resist attacks from quantum computers, offering robust security under the random oracle model [3]. Considering the best known quantum algorithm subject to break PQ cryptography nowadays – Grover’s algorithm – ML-KEM-1024 provides 256 bits of strength. On the other hand, QKD provides an inherently secure environment that is resistant to any exhaustive key search or cryptanalytic attack, even those originating from quantum computers [29].

Thus, we can assume that the combination of PQ cryptography and QKD will provide as a result a hybrid shared secret backed by two distinct quantum-resistant cryptographic assumptions, achieving at least 256 bits of security. Furthermore, if QKD is implemented properly, the final shared secret also becomes ITS secure.

Such composite solutions provide a fallback mechanism in case one of the cryptographic assumptions is broken in the future. A literature study of the security proofs of triple-hybrid key combiners involving classical, PQ cryptography, and QKD can be found at [33], [34], [35].

For authentication, impersonation cannot happen retroactively; therefore, contrary to encryption – which is already vulnerable to HNDL attacks – authentication can still be addressed via classical cryptography, at least until a powerful-enough quantum computer is available. Nonetheless, NIST has already established a deadline by 2035 for communication systems to implement quantum-resistant authentication, underscoring the importance of a timely adoption [36].

Building on these assumptions, we envision that the combination of both quantum-resistant technologies will help, for example, to alleviate the existing fear towards already adopting PQ or QKD-based communications. Furthermore, the crypto-agility of our system allows one to immediately replace any cryptographic algorithm that might be subject to a future algorithmic breakthrough or, on the other hand, to add additional cryptographic techniques that enhance the security of the AKE procedure. With this, our approach provides ITS, forward and post-compromise security in the current landscape of quantum computing, immediately addressing HNDL attacks [34].

IV. QUANTUM-RESISTANT TLS 1.3

A. Transport Layer Security 1.3

TLS 1.3 is a network security protocol designed to provide secure communications over the Internet between two communicating parties according to RFC 8446 [7]. It builds on top of a reliable transport layer protocol such as the transmission control protocol (TCP) [37], and it requires that the secure channel established by TLS should provide authentication, confidentiality and integrity [7].

TLS accomplishes these security goals through two sub-protocols: the TLS handshake and the TLS record. The handshake protocol is responsible for authentication, key

negotiation and key exchange, utilizing PKC operations such as digital signatures, ephemeral key-pair generation, and key encapsulation/decapsulation [29]. Currently, these PKC operations are implemented using classical cryptography algorithms, such as RSA and EC cryptography, as defined in the RFC 8446 standard [7]. However, as explained in Section II, TLS necessitates a migration of all PKC cryptographic operations towards quantum-resistant technologies, such as PQ cryptography and QKD.

In contrast, the record protocol uses the negotiated secret keys from the handshake to implement symmetric encryption (e.g., AES-256-GCM) for secure data transmission. Although quantum search algorithms like Grover’s theoretically pose a threat to symmetric encryption, their efficiency can be mitigated through key expansion. As a result, AES-GCM with sufficiently large keys is expected to remain quantum-resistant in the upcoming future, assuming that no new quantum algorithm is created to offer a speedup significantly larger than that from Grover’s algorithm. Therefore, the TLS record protocol can be considered quantum-resistant if implemented via an appropriate quantum-resistant AEAD symmetric encryption algorithm such as AES-256-GCM [11], [38].

To achieve full quantum resistance, it is essential to migrate the TLS handshake protocol towards using quantum-resistant methods for authentication and key exchange. For this, there are two clear options. PQ cryptography can be used for both authentication and key exchange, as it replicates the behavior of traditional PKC algorithms. QKD can only be applied towards implementing key exchange due to the nature of the underlying operations to obtain a shared secret key at both endpoints of the communications.

Thus, our work proposes the following. To employ either classical or PQ cryptography for authenticating parties. Authentication cannot happen retroactively, and thus, it is not an urgent problem, at least until a powerful enough quantum computer becomes available [20]. A sample implementation of the TLS 1.3 protocol with classical authentication and hybrid quantum-resistant AKE is shown in [39]. Nevertheless, it is well known that the transition towards quantum-resistant authentication must happen sooner than later. Again, our work in [20] shows a version of our quantum-resistant TLS 1.3 protocol both implementing PQ-only quantum-resistant authentication and hybrid quantum-resistant key exchange.

Both alternatives would work nowadays, with classical cryptography being the preferred option today because: 1) the maturity of the algorithms and their security analyses; and 2) message sizes are more suitable than PQ cryptography. However, PQ authentication is the preferred option to protect against known quantum algorithms. For the current work, and following the latest standards for PQ digital signatures [40], we have chosen to implement CRYSTALS-Dilithium, also known as ML-DSA-65 for digital signatures and digital signature verification.

Regarding AKE, we follow the Dutch national recommendations as per [38] and implement a hybrid version of TLS that uses classical EC Diffie-Hellman (DH) ephemeral (ECDHE) in combination with PQ-based ML-KEM, formerly known as CRYSTALS-Kyber [19], which is the PQ key encapsulation

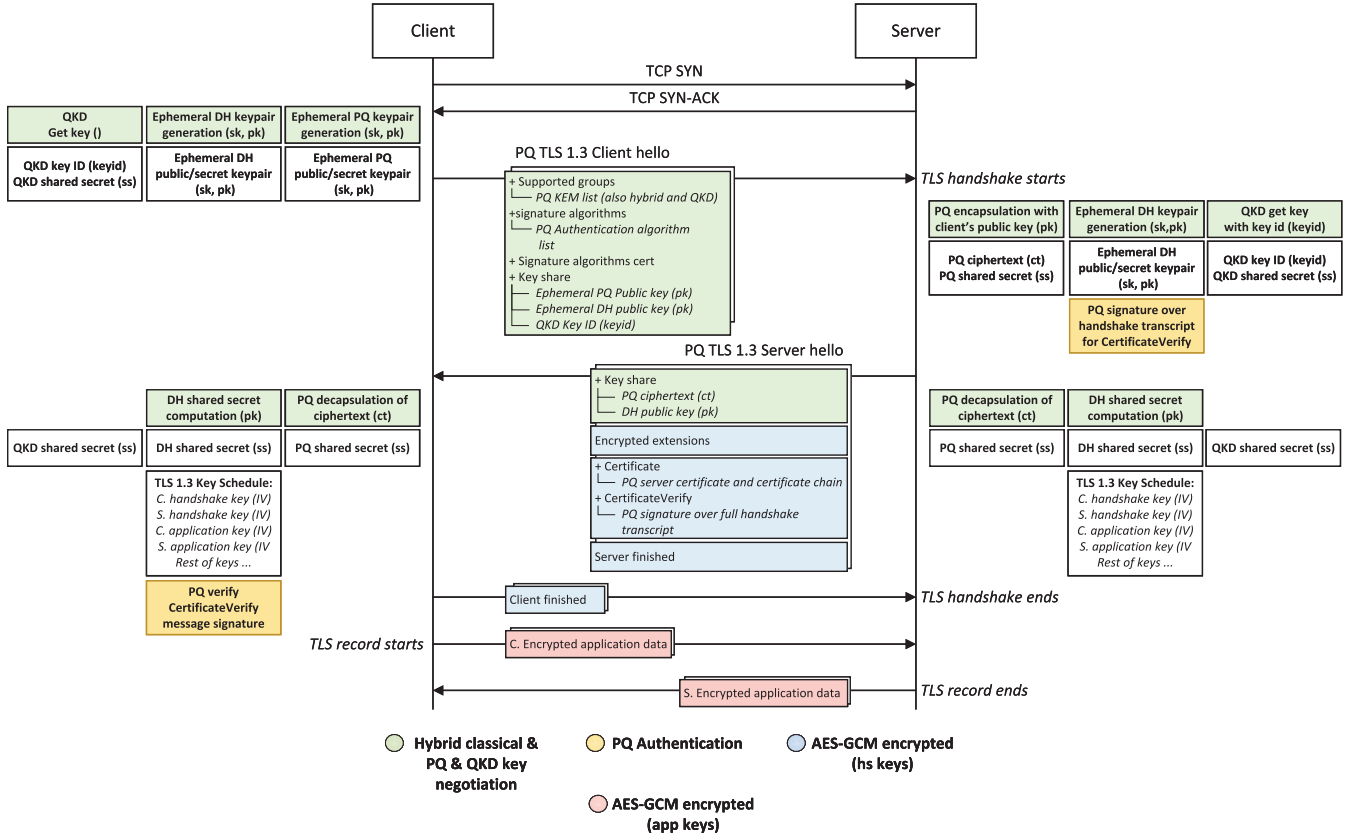


Fig. 2. Handshake workflow for integrating classical, PQ cryptography and QKD into our hybrid quantum-resistant TLS 1.3.

mechanism selected by NIST as the preferred standard [18]. As recommended, we use the parameter set ML-KEM-1024. Moreover, we also test other PQ-KEMs proposed in [38] for benchmarking purposes. Finally, we have included our custom libraries for integrating the QKD key material into the handshake, thus complementing the aforementioned cryptography techniques and achieving triple-hybrid security.

It must be said that, compared to our previous work in [20], we have modified the TLS 1.3 codebase so that every possible key negotiation, including the QKD one, is now triggered at the client's side before the `client hello` message, as shown in Fig. 2. This ensures both that, as mentioned in [29], no key exhaustion attacks at the quantum layer can take place on the server, and that we can achieve performance improvements compared to the work in [20].

As described by current standards for hybridization [41], there are two different ways in which shared secrets generated by several cryptographic assumptions can be combined to achieve hybrid structures: concatenate and cascade approaches. Our implementations utilizes the concatenation-based approach, as it allows a very agile introduction of new key exchange mechanisms into the TLS key schedule without major modifications. In our work, the resultant input key material (IKM) concatenates three shared secrets coming from classical, PQ cryptography, and QKD into a single byte array, which is fed to the TLS key schedule where key derivation takes place, and where all the needed cryptographic key material is derived [7].

The solution we present in this work integrates QKD in TLS 1.3 through OpenSSL, Liboqs, and our custom developed libraries for QKD [7], [10], [42]. We extended these libraries to integrate QKD for the following reasons: 1) OpenSSL is one of the most widely used TLS libraries both in industry and academia, thus ensuring plug-and-play integration across existing infrastructure and; 2) Liboqs is a well known open source C library for PQ cryptography developed by the Open Quantum Safe team [43] and provides compatibility with PQ algorithms as well as classical cryptography.

By building our solution on top of libraries written in C programming language, we can ensure that our solution is suitable for a wide variety of network elements, communications devices or computing machines that would be deployed in current telecommunications infrastructures. Moreover, the interoperability of the libraries ensures that cryptographic algorithms can be chosen at runtime, facilitating an easy and agile transition from classical, PQ crypto, QKD and vice versa, and the deployment of hybrid and triple-hybrid quantum-resistant communication protocols depending on the user requirements.

B. Key Derivation: Tls 1.3 Key Schedule

The key schedule is a cryptographic procedure that utilizes a hash-based key derivation function (HKDF) to derive the key material [44]. The HKDF is a cryptographic procedure used in various network security protocols and applications, and whose main objective is to derive cryptographically strong keys from a source of IKM that may not be uniformly random.

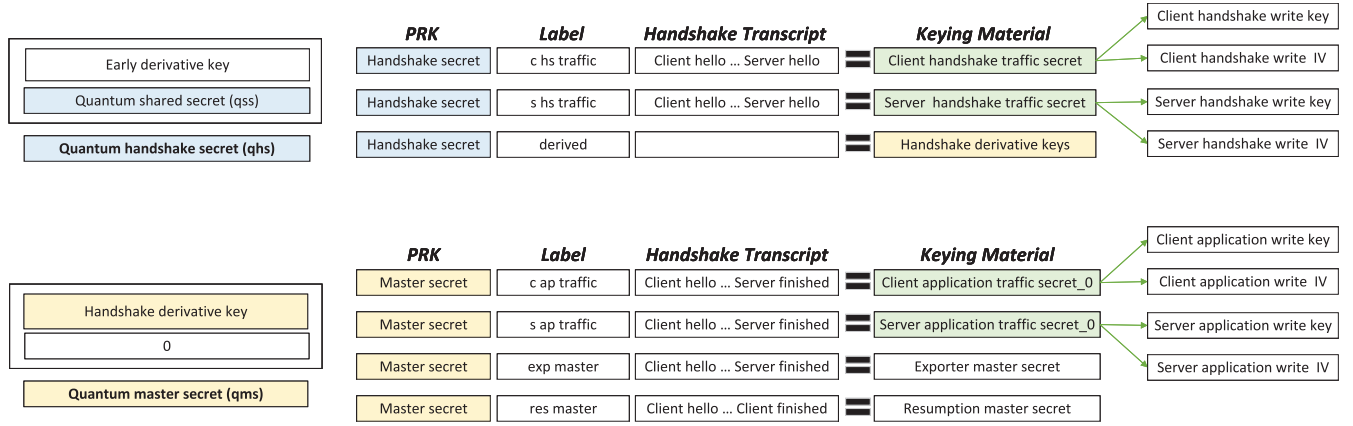


Fig. 3. TLS 1.3 key schedule procedure for deriving handshake and application write (encryption) keys [45]. Step 1) describes the procedure for deriving handshake encryption keys and IKM for application encryption keys. Step 2) describes the procedure to obtain application encryption keys to be used by client and server to encrypt traffic data.

The HKDF happens symmetrically at both the client and server endpoints, following an ‘extract-then-expand’ methodology consisting of two different stages. The ‘extract’ stage takes the source IKM and concentrates the possibly dispersed entropy into a short, cryptographically strong pseudorandom key (PRK). The second stage is known as the ‘expand’ stage, and it expands the PRK to a byte array of the desired length, commonly known as the output key material (OKM), which can later on be separated into several additional PRKs that are ready to use.

In the case of TLS 1.3, as shown in Fig. 3, a set of seven secrets (e.g., Client Handshake Traffic Secret, Server Handshake Traffic Secret, Handshake derivative keys, Master Secret, Client Application Traffic Secret, Server Application Traffic Secret, Exporter Master Secret and Resumption Master Secret) are derived from the initial IKM provided by either classical, PQ crypto, QKD, or all of them combined. The IKM is used as the seed for deriving the different encryption keys to be utilized in the protocol [7].

For the present work, the IKM is a 96-byte array formed by three shared secrets – classical, PQ and QKD – each of them composed by 32 bytes, and specified by us as the quantum handshake secret (QHS). As a result, the security of the resultant PRKs is guaranteed by three cryptographic assumptions, each one independent of each other and quantum-resilient. Nevertheless, it is worth mentioning that the current implementation allows the combination of multiple cryptographic protocols as the IKM, since the length of the OKM is fixed at the ‘extract’ phase to 32 bytes. This allows some degree of customization depending on the security requirements of each user.

During the key schedule derivation process described in Fig. 3, the QHS serves as the IKM to the ‘extract’ procedure discussed above. The procedure derives a set of handshake secrets and handshake derivative keys. The first two are then employed, via another ‘extract’ mechanism similar to the one discussed above, to derive a set of handshake write (encryption) keys and their respective initialization vectors (IV) which are used to encrypt the handshake messages.

The latter (handshake derivative key) – namely quantum master secret (QMS) – is then used again as the IKM for a

second ‘extract-then-expand’ procedure, from which a set of application traffic secrets are derived.

These application traffic secrets are then used to generate the final application write keys and IVs, which are the secret keys to encrypt the final application data exchanged among users. This methodology allows to isolate the secret keys used to encrypt the application data (application write keys) from the rest of the keys used to encrypt the handshake messages (handshake write keys), providing an extra layer of security.

Finally, all derived write keys, whether used to encrypt handshake messages or application data, can be used by TLS via a quantum-resistant AEAD symmetric encryption algorithm such as AES-256-GCM or ChaCha20-Poly1305. These symmetric encryption algorithms are considered quantum-resistant, and thus, can be used for the exchange of encrypted application data between client and server [38].

A well known challenge of QKD implementations is the SKR limitations intrinsic to the short achievable physical distance among nodes. These are imposed by both the distance-SKR trade-off explained in Section II and the availability of keys at the KMS layer that the SAEs require.

In this regard, the TLS 1.3 key schedule procedure proposes the renewal of session keys at the application traffic secrets level, depicted as the bottom layer of Fig. 3. This implies that, independently of the encrypted traffic volume that is encrypted, no need for new key requests to the KMS layer are required. Overall, only one shared secret key must be employed to trigger and maintain a TLS session among client and server, which eases the integration of QKD due to the low key requirements of the protocol.

In our specific TLS 1.3 implementation, we consider our current QKD equipment’s SKR of approximately 2.5 kbit/s and the size of the QKD-based key equal to 256 bits. Using Eq. (1), we can estimate that our current proposal can secure up to 78 new high-speed TLS user connections (UC) per second [20].

$$UC = \frac{QKD\ SKR\ (bits/s)}{Key\ Size\ (bits)} \quad (1)$$

V. QUANTUM-RESISTANT IPSEC

A. Internet Protocol Security

Internet protocol security (IPsec) is a network security protocol designed to secure communications within the Internet Protocol (IP). Defined in RFC 4301, IPsec operates at OSI layer 3 and provides cryptography-based security services for IPv4 and IPv6. It ensures confidentiality, data integrity, access control, and authentication of IP datagrams across both control and data planes [8], [29].

In the IPsec control plane, the Internet key protocol version 2 (IKEv2) is used to negotiate, among others, the cryptographic algorithms and keys that will be used to encrypt authentication header (AH) and the encapsulating security payload (ESP) packets [46]. Although the security of AH and ESP packets is based on symmetric cryptography (i.e., AES), the IKEv2 protocol relies on PKC to negotiate a secret that can be securely shared by the initiator and responder. Thus, the IKEv2 protocol is vulnerable to Shor's algorithm and necessitates a transition to quantum-resistant cryptography.

In the data plane, most of the security services employed for encryption are provided via ESP and AH, which protect data sent through an IPsec security association (SA) [47]. To enable quantum-resistant symmetric encryption in the data plane, a successful and proven approach is to employ AES-256-GCM [38], [48].

B. Internet Key Exchange Version 2

The IKEv2, defined in RFC 7296 [46], is a sub-component of the IPsec protocol used for performing mutual authentication and establishing and maintaining SAs. It negotiates security parameters among initiator and responder (e.g., client and server), performs mutual authentication, and establishes the SAs, including the shared secret keys that are necessary for authentication headers (AH) and encapsulating security payloads (ESP).

In our case, we have modified the protocol to become quantum-resistant by: 1) Allowing only quantum-resistant AEAD such as AES-256-GCM; 2) adding an additional exchange to negotiate a PQ-based shared secret; 3) adding an additional exchange to negotiate a QKD-based shared secret; 4) integrating all quantum-resistant cryptographic techniques via RFC 9370, rather than using pre-shared keys [49], [50].

Usually, as seen both in academia and industry, the integration of QKD into IPsec has been carried out by leveraging the use of post-quantum pre-shared keys (PPKs) mechanisms defined in RFC 8784 [50]. Examples of such implementations leveraging PPKs can be found on [51], [52], [53], and [54]. Nonetheless, the PPK approach comes with certain disadvantages, which are described below and classified in Table I.

First, RFC 8784 (2020) offers a short-term solution using post-quantum pre-shared keys (PPKs), while RFC 9370 (2023) provides a comprehensive long-term quantum-resistant upgrade to the IKEv2 protocol itself to support PQ algorithms [49], [50].

Regarding the procedure in which shared secrets are transmitted between the initiator and responder, RFC 8784 assumes that both PPKs and their corresponding key identifiers have already been securely exchanged via an ad hoc channel. RFC

TABLE I

COMPARISON OF RFC 9370 [49] AND RFC 8784 [50] FOR IMPLEMENTING QUANTUM-RESISTANT IKEV2

	RFC 9370	RFC 8784
Release Date	2023	2020
Support	Long-term solution	Short-term extension
Key negotiation	Integrated	Ad hoc, PPKs
Rekeying	Yes	No
Scalability	Yes	No
Hybrid Schemes	Multiple	Single
Crypto-agility	Yes	No
Control Plane Security	Full	Limited
Related Work	-	[51]–[54]

9370 proposes rather mechanisms in which the quantum-resistant key negotiation is fully integrated into the IKEv2 protocol, similar to how classical algorithms are handled.

Regarding rekeying, RFC 8784 only specifies how the PPK is used only during the initial IKE SA setup. On the other hand, RFC 9370 also defines a mechanism through which the shared secret material of subsequent SAs can be refreshed, providing fresh sources of entropy to the key material.

Furthermore, RFC 8784 requires unique PPKs for each peer pair, limiting scalability [50]. In contrast, RFC 9370 negotiates shared secrets in a dynamic way, fully integrated into the IKEv2 protocol, making the key negotiation procedure much more scalable.

Concerning the issue that some of the PQ algorithms might turn out easier to break than anticipated, RFC 9370 allows to integrate more than one quantum-resistant algorithm, allowing for double, triple, or even larger hybrid key exchanges, while RFC 8784 only allows for a single PPK.

Finally, as stated in RFC 8784, the mechanism in which the PPK is integrated into the IKEv2 key derivation procedure does not target the initial IKE SA, but rather the subsequent IKE SAs created after the initial IKE SA is rekeyed. RFC 9370 provides full control plane security by negotiating quantum-resistant algorithms before establishing the initial IKE SA, addressing potential quantum attack vulnerabilities inherent in RFC 8784.

Considering the numerous factors summarized in Table I, we consider RFC 9370 as the best choice for implementing a quantum-resistant IKEv2 protocol. Moreover, the crypto-agility intrinsic to RFC 9370 ensures that implementations of the IKEv2 protocol can adapt to emerging security threats. Therefore, to the best of the author's knowledge, we present the first implementation of a triple-hybrid IKEv2 protocol seamlessly integrating classical and PQ cryptography, and QKD into the IKEv2 via RFC 9370.

In our implementation, each IKEv2 peer performs multiple successive key exchanges to establish an IKE SA. The output of each key exchange, namely a shared secret, is then combined in a 'cascade' approach such that: 1) The final hybrid shared secret is derived from all component key exchanges; 2) the format of the final hybrid shared secret is equivalent to the final shared secret specified in legacy IKEv2 key negotiation

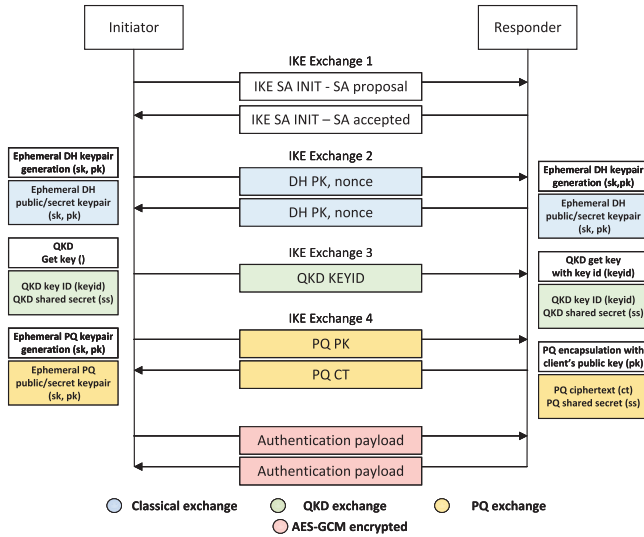


Fig. 4. Multiple Key Exchanges in the Internet Key Exchange Protocol Version 2 (IKEv2): Combining hybrid classical, PQ cryptography and QKD-based IKEv2 in RFC 9370 [49].

procedures [46]; 3) If any component of the key exchange procedure is considered PQ secure, then the final hybrid shared secret is PQ secure; 4) If QKD is used integrated in the key exchange procedure, the final hybrid shared secret becomes ITS secure.

In the context of quantum-resistant IKE, the message sequences depicted in Fig. 4 are required to agree on the parameters to be employed for encrypting communications. The IKE SA INITexchange negotiates the cryptographic algorithms, shares nonces, signals the need for extra key agreements, and performs a DH key exchange. Additionally, as defined in RFC 9370, it informs the protocol that following the IKE SA INITexchange, a series of successive key exchanges will take place via IKE INTERMEDIATEexchanges to negotiate a quantum-resistant shared secret.

For each additional key exchange agreed during the initial IKE SA INITexchange, the initiator and responder must perform an extra IKE INTERMEDIATEexchange, where the relevant cryptographic parameters are exchanged so that a shared secret can be securely derived on both sides. As this exchange occurs after the first cryptographic secret is negotiated, the message is protected with the current SK_{ei}/SK_{ai} keys negotiated during the IKE SA INITexchange. With PQ algorithms, the IKE INTERMEDIATEmessage contains either a public key or a ciphertext. With QKD, only the key ID is exchanged. These are public parameters and thus, although they benefit of classical confidentiality, they are not expected to be secret. As a result, after the IKE SA INITand IKE INTERMEDIATEexchanges have occurred, the following cryptographic functions have been negotiated among the initiator and the responder: 1) an encryption algorithm; 2) an integrity protection algorithm; 3) a DH group; 4) hybrid algorithms providing quantum-resiliency; 5) a PRF.

The third exchange, namely IKE AUTH, authenticates the previous messages, exchanges identities and certifi-

cates, and establishes the first child SA. Subsequent IKE exchanges for communication and rekeying are called either CREATE CHILD SAor INFORMATIONALexchanges. CREATE CHILD SAexchanges allow to create additional child SAs, as well as rekey already on-going IKE and child SAs. INFORMATIONALexchanges are utilized for deleting SAs, reporting errors, or other maintenance tasks.

In our work, as depicted in Fig 4, we introduce two additional quantum-resistant key exchanges on top of the classical key exchange. The first extra exchange corresponds to the negotiation of a QKD-based shared secret. The second exchange negotiates the use of a PQ-based shared secret, namely ML-KEM-1024. Overall, the sizes of the messages required to establish a QKD key exchange among initiator and responder (e.g., 36 bytes) allows us to do it with only one extra exchange. For PQ-based alternatives such as ML-KEM, depending on the parameter set negotiated, more than one additional exchange might be needed, due to the much longer public key and ciphertext sizes.

C. Key Derivation: IPsec Key Mixing in IKEv2

The key derivation process in IPsec is based on the security parameters negotiated during the IKEv2 protocols explained in Section V-A. Overall, IKEv2 negotiates the security parameters for two separate SA tunnels: IKE SAs and child SAs.

The IKE SA tunnel is used as the control plane tunnel or the management layer. It is responsible for setting up, maintaining, and tearing down secure connections through child SAs. Child SA tunnels are the actual data plane tunnels that end-users employ to exchange confidential data. Each tunnel owns a specific set of cryptographic keys that are used for encrypting communications and that must be renewed each specific amount of time.

1) *IKE SAs Key Derivation*: In the initial IKE SA, The shared secret keys are computed in a two-step procedure as follows. First, a key called *keyseed* is calculated via Eq. (2) by using the nonces exchanged during the initial IKE SA INITexchange and the initial DH shared secret [46]. The *keyseed* takes as input the initiator's nonce (N_i), the responder's nonce (N_r), and a 32-byte shared secret (g_{ir}) established via DH.

$$\text{keyseed} = \text{prf}(N_i \mid N_r, g_{ir}) \quad (2)$$

Secondly, the negotiated *keyseed* is then used to calculate seven other secrets, namely *keymat*, shown in Eq. (3) and Eq. (4) [46]. SK_d is used to derive new keys for child SAs within the current IKE SA; SK_{ai} and SK_r are used as input for the integrity protection algorithm to authenticate subsequent exchanges; SK_{ei} and SK_{er} are used for encryption and decryption of all subsequent exchanges; SK_{pi} and SK_{pr} are used to generate and verify authentication payloads. As can be seen in Eq. (4), the *keymat* is finally sequentially divided into the final key blocks $SK_d, SK_{ai}, SK_{ar}, SK_{ei}, SK_{er}, SK_{pi}, SK_{pr}$ which serve to encrypt application data.

$$\text{prf}^+(\text{keyseed}, N_i \mid N_r \mid SPI_i \mid SPI_r) = \text{keymat} \quad (3)$$

keymat

$$= \{SK_d \mid SK_{ai} \mid SK_{ar} \mid SK_{ei} \mid SK_{er} \mid SK_{pi} \mid SK_{pr}\} \quad (4)$$

The procedure is then repeated as many times as key exchanges have being negotiated during the IKE SA INITExchange. During each iteration, the new *keyseed* is derived in a ‘cascade’ approach via a PRF, which uses the previously negotiated shared secret $SK_d(N-1)$, the newly negotiated shared secret $SK(n)$, and the initiator’s (N_i) and responder’s nonces (N_r) exchanged during the initial IKE SA INITExchange, as shown in Eq. (5) [49]. Similarly to how it happens in the initial key exchange, the newly negotiated *keyseed*(n) is then used again in Eq. (3) to update the rest of the key material.

$$\text{keyseed}(n) = \text{prf}(SK_d(n-1), SK(n), |N_i \mid N_r) \quad (5)$$

Finally, the key blocks described in Eq. (4) are utilized to encrypt communications during the initial child SA establishment procedure.

2) *Child SAs Key Derivation*: The CREATE CHILD SAexchanges are used in IKEv2 for the purposes of creating additional child SAs, rekeying already on-going child SAs, and rekeying IKE SAs. When creating or rekeying child SAs, peers might optionally choose to perform an additional key exchange to add fresh entropy to the session keys – this is the approach we use in our work – or to reuse the cryptographic key material negotiated before. In our presented implementation, we use the initial IKE SA key material as the seed to derive the *keymat* of the initial child SA. For subsequent exchanges, and in order to guarantee the use of fresh entropy, CREATE CHILD SAexchanges are used to establish new child SAs for which fresh key material is used as the IKM.

Similarly to how secret key material is negotiated during the initial IKE exchange, additional quantum-resistant key exchanges can be negotiated to ensure that the *keymat* of child SAs remain quantum-resistant. However, the procedure is somewhat different to the one used for IKE SAs. During the procedure of creating or rekeying a child SA, additional key exchanges are negotiated via IKE FOLLOWUP KEexchanges. Rather than updating the key material at each iteration, all additional cryptographic key material is fed into a PRF+, in a ‘concatenation’ approach, as shown in Eq. (6). The output of the PRF+ is a byte array of a specific length, from which the rest of the traffic keys can be extracted. After the *keymat* is derived, according to the specific length requirements of the underlying encryption algorithms, the same procedure shown in Eq. (4) is used to divide *keymat* into seven different secret key blocks.

$$\text{keymat} = \text{prf}^+(SK_d, SK(0)|N_i \mid N_r|SK(1)|\dots|SK(n)) \quad (6)$$

In the case of an IKE SA rekey, CREATE CHILD SAexchanges are also used. In that case, the resulting keys are computed as seen in Eq. (7), where the output of the PRF is the *keyseed* that is used to derive the rest of the key material.

$$\text{keyseed} = \text{prf}(SK_d, SK(0)|N_i \mid N_r|SK(1)|\dots|SK(n)) \quad (7)$$

In any case where a CREATE CHILD SAexchange is employed, whether to create a child SA, or to rekey an IKE SA or child SA, SK_d comes from the existing IKE SA; $SK(0)$, N_i and N_r are the shared keys and nonces negotiated during the CREATE CHILD SA. The subsequent shared key materials $SK(1), SK(2) \dots SK(n)$ are those negotiated during the IKE FOLLOWUP KEexchanges.

3) *Pseudorandom Functions*: As can be seen above in Sections V-C1 and V-C2, the core of the key derivation operations within the IKEv2 protocol is based on two different cryptographic functions. PRF and PRF+. As shown in Eq. (8), a PRF is a cryptographic function that takes a secret key, commonly known as the IKM, a seed, which represents a variable input to the PRF (i.e., concatenated elements, nonces, labels, etc.) and produces an OKM of fixed length that appears to be random to an adversary. If the IKM is assumed to be cryptographically secure, then the OKM inherits the security assumptions of the IKM.

$$\text{prf}(\text{ikm}, \text{seed}) = \text{OKM} \quad (8)$$

In cases where the total required key material is larger than the fixed output length of the PRF, such as when negotiating the total *keymat* needed by IKEv2, the PRF can be used iteratively until all the necessary key material has been derived. This is what is known as PRF+. A PRF+ is an extension of a PRF used to generate a pseudorandom stream of data that can be longer than the fixed output length of a PRF. The PRF+ works by concatenating the outputs of successive PRF invocations, as shown in Eq. (9).

$$\begin{aligned} \text{prf}^+(K, S) &= T_1 \parallel T_2 \parallel T_3 \parallel \dots \\ T_1 &= \text{prf}(K, S \parallel 0 \times 01) \\ T_2 &= \text{prf}(K, T_1 \parallel S \parallel 0 \times 02) \\ T_3 &= \text{prf}(K, T_2 \parallel S \parallel 0 \times 03) \end{aligned} \quad (9)$$

D. Key Renewal Intervals

Regarding the key renewal time interval of the cryptographic key material, considering that quantum-resistant encryption is implemented through AES-256-GCM, the theoretical safety limit for key usage is set at 379 GB of encrypted data using the same key [7]. Taking this information into account, as well as considering the implementation of a high-throughput IPsec tunnel (100 Gbps), we can use Eq. (10) shown in [29] to estimate that, for each child SA (tunnel), the SK_d , and thus the *keyseed* secret should be renewed each 30 s. With our current equipment, based on ID Quantique Clavis3 equipment [55], our average SKR 2.5 kbit/s translates into 9 AES-256-GCM-enabled keys (256 bits) per second, which fits the requirements in terms of key material available and high-speed communications to create high-throughput quantum-resistant IPsec tunnels.

$$\text{Renewal Interval (s)} = \frac{\text{Key usage limit (GB)}}{\text{High-speed connection (Gb/s)}} \quad (10)$$

This approach allows to establish secret key material protected by three different cryptographic assumptions, and provides a high degree of crypto-agility that allows the presented solution to quickly switch to a different combination of

cryptographic algorithms considered quantum-resistant in the face of any possible cryptanalytic attack, technology discovery, or algorithmic breakthrough.

VI. RESULTS AND DISCUSSION

In this section, we describe the experimental setup employed to analyze our quantum-resistant TLS and IPsec solutions, including both the hardware and software elements involved. We also present and discuss the results obtained from our own Eindhoven QKD testbed [56].

A. Experimental Setup

Our experimental setup, for both TLS 1.3 and IPsec implementations, involves two QKD nodes connected through optical fibers in a point-to-point connection. Each QKD node is composed of the following: 1) a QKD module; 2) a KMS; and 3) a quantum-resistant application (e.g., TLS, IPsec client/server). Our quantum-resistant applications, whether TLS (layer 4) or IPsec (layer 3), are hosted in Ubuntu-based docker containers (Ubuntu 22.04 LTS) located in two separate servers, each corresponding to a QKD node. Regarding the QKD modules, our testbed integrates the Clavis3 equipment from IDQuantique. Both QKD modules are directly connected through a dark fiber serving as the ‘quantum channel’, facilitating the exchange of keys at an average SKR of 2.5 kbit/s. Regarding security, the Clavis3 system operates using the Coherent One-Way (COW) protocol, which is part of the DV-QKD family. The COW protocol relies on encoding information into coherent pulses of light sent over an optical fiber and therefore uses the principles of quantum mechanics to achieve security. A detailed analysis of the COW protocol is found in [57].

Finally, the rest of the equipment is connected via a ‘classical channel’ through a network switch, enabling classical communications. Through this classical channel, and by retrieving the keys that have been already transmitted at the quantum layer, our TLS and IPsec applications use the quantum shared secrets to implement quantum-resistant symmetric encryption (AES-256-GCM). Regarding the software elements, our solution builds on the latest versions of OpenSSL3.4.0 [10], strongSwan6.0.6 [9], and Liboqs0.10.0 [42], [43].

B. Discussion and Results on Quantum-Resistant Tls 1.3

While the main motivation for our work is to analyze a TLS 1.3 implementation where three different cryptographic assumptions are integrated into the handshake, thus achieving enhanced quantum-resiliency, the communication cost of integrating such triple-hybrid alternatives is an important aspect to take into account. In our previous work shown in [39], we evaluate several options for single and hybrid cryptographic combinations. Furthermore, the work presented in [20] studies the use of triple-hybrid cryptographic algorithms, but triggers the QKD key retrieval process at the server’s side before the `client hello` message is exchanged. Although secure, such implementation was subject to possible key exhaustion attacks on the server.

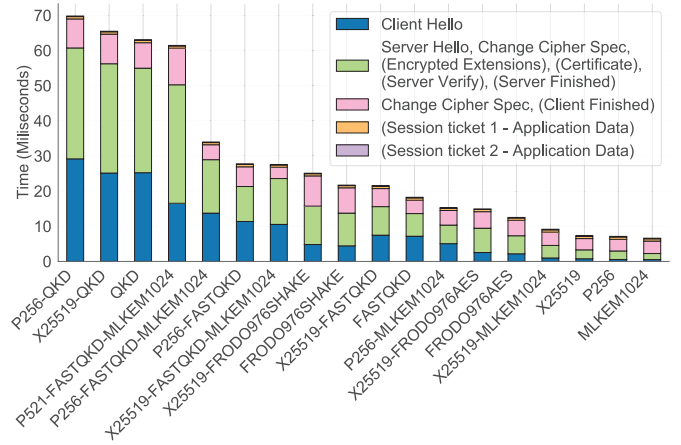


Fig. 5. TLS 1.3 timings (per outbound message) using various cryptographic (quantum-resistant) algorithms in a single, hybrid and triple-hybrid mode.

Having into account the aforementioned issue, as well as the need for triple-hybrid security schemes, the presented solution goes a step further and enables the use of triple-hybrid cryptographic combinations where every possible key negotiation technique, whether classical, PQ or QKD, is triggered at the client’s side before the first `client hello` message is exchanged. The technical complexity of developing such a solution (e.g., runtime management, interoperability, scalability) minimally impacts the total handshake time compared to the previous, more simple implementation shown in [20]. Nevertheless, the performance gain given by already triggering the QKD key exchange at the client’s side compensates the added technical complexity, even outperforming the work previously shown in [20]. Nonetheless, the developed handshake architecture ensures that no key exhaustion attacks to the quantum layer can take place on the server. Furthermore, as well as minimally improving the performance of the protocol and enhancing the security, the presented architecture opens new possibilities for accelerating the triple-hybrid key negotiation procedure, as well as to implement a different kind of triple-hybrid mechanisms based on two different PQ algorithms (e.g., ML-KEM-1024+ FrodoKEM976).

For benchmarking our implementation, we use the classical algorithm X25519 and the PQ algorithm ML-KEM-1024 as control algorithms. For benchmarking state of the art hybrid algorithms based on classical and PQ cryptography, we have chosen the combination of X25519 and ML-KEM-1024. Regarding authentication, this TLS implementation employs ML-DSA-65. As it can be seen in both Fig. 5 and table II, there is a significant difference in two different steps of the handshake with respect to the others – `client hello`, and `server hello` messages.

Client hello. Before the `client hello` can be sent from client to server, for both classical and PQ cryptography, a key-pair generation needs to take place in order to generate a private-public keypair. Implementations that use ML-KEM-1024 show a very similar performance to the classical algorithm X25519, even outperforming it. Regarding hybrid combinations based on X25519 and ML-KEM-1024,

TABLE II
 TLS 1.3 TIMES (PER HANDSHAKE STEP) WITH DIFFERENT CRYPTOGRAPHIC (QUANTUM-RESISTANT) ALGORITHMS.
 RESULTS ARE PRESENTED IN MILLISECONDS (MS)

Algorithm	Client Hello	ServerHello, ServerCCS, EncryptedExtensions Certificate ServerVerify, ServerFinished	ClientCCS, ClientFinished	Session Ticket 1	Session Ticket 2	Handshake time
P256-QKD	29.18	31.55	8.25	0.67	0.16	69.80
X25519-QKD	25.14	31.10	8.42	0.66	0.15	65.47
QKD	25.24	29.73	7.25	0.67	0.15	63.04
P521-FASTQKD-MLKEM1024	17.09	33.12	11.25	0.61	0.17	62.24
P256-FASTQKD-MLKEM1024	13.79	15.08	5.06	0.61	0.16	34.69
X25519-FASTQKD-MLKEM1024	10.57	12.85	3.67	0.62	0.17	27.87
P256-FASTQKD	11.37	9.94	5.61	0.67	0.17	27.76
X25519-FRODO976SHAKE	4.81	10.93	8.59	0.62	0.14	25.09
FRODO976SHAKE	4.44	9.31	7.20	0.62	0.14	21.70
X25519-FASTQKD	7.46	8.13	5.13	0.66	0.18	21.55
FASTQKD	7.17	6.45	3.79	0.67	0.17	18.24
P256-MLKEM1024	5.07	5.27	4.16	0.64	0.17	15.29
X25519-FRODO976AES	2.54	6.91	4.71	0.63	0.15	14.93
FRODO976AES	2.17	5.13	4.39	0.65	0.16	12.52
X25519-MLKEM1024	0.98	3.57	3.78	0.65	0.17	9.15
X25519	0.76	2.53	3.21	0.65	0.18	7.33
P256	0.56	2.42	3.31	0.67	0.17	7.12
MLKEM1024	0.53	1.72	3.54	0.64	0.17	6.61

the total time to perform the handshake remains low as both cryptographic alternatives perform fairly well.

Regarding implementations involving QKD, we can observe a significant difference with respect to the other cryptographic techniques presented. This is mainly due to the performance of the application programming interface (API) employed by our KMS's to safely deliver keys to the requesting SAEs [27]. As explained in Section II, this is one of the main challenges of QKD in terms of performance. For instance, Fig. 5 shows that algorithms involving "QKD" perform considerably slower, with an added overhead of approximately 24 ms each time a key is delivered. To show how a fast API implementation would perform in a triple-hybrid scenario, we also consider a emulated "FASTQKD" implementation, whose key retrieval procedures takes approximately 10 ms per API call.

While we have previously shown results of our triple-hybrid encryption in [20] involving the aforementioned algorithms, we have made significant security and performance advancements in the development of an in-house KMS, as well as the new TLS architecture shown in Fig. 2 for integrating triple-hybrid algorithms, which now triggers all the cryptographic techniques before the `client hello` message is exchanged. Thus, for a triple-hybrid implementation of TLS protocol based on X25519, ML-KEM-1024 and QKD, and thanks to the high key generation performance of both classical and PQ algorithms, we can conclude that the total overhead added before the `client hello` message is sent mainly depends on the performance of the QKD key delivery procedure.

Server hello. After the `client hello` message has been processed on the server side, there are three cryptographic messages as outcome: 1) Client's DH public key; 2) The client's ML-KEM-1024 public key; and 3) A QKD key ID. Implementations using classical DH require of another key-pair generation operation on the server's side, as the server's

public key must be again transmitted back to the client to perform the key derivation process. Regarding PQ, since all of the options are KEM's, a key encapsulation takes place, where the server generates a shared secret which is encrypted and sends back the ciphertext to the client. Finally, the key ID can be used by the server to retrieve the exact same quantum key as the client, which has previously being exchanged safely across the quantum layer. As it happened in the `client hello` message, the performance of key retrieval has margin for improvement, and has a performance impact of approximately 24 ms over the overall handshake time.

Thus, the total overhead added by QKD is of approximately 48 ms by using real equipment. This is certainly a challenge of QKD that must be further studied. Achieving key retrieval times of approximately 10 ms would greatly enhance the usability of QKD equipment in real time encryption protocols. In terms of message size overhead, as shown in Fig. 6, one of the key advantages of integrating QKD into network security protocols is its low bandwidth requirements. With only 36 bytes required for the key identifier, QKD can be incorporated into the handshake with negligible impact on packet overhead. Thus, in terms of message size overhead, it is PQ cryptography the primitive that has a biggest impact on packet sizes.

Finally, it must be noted that the TLS 1.3 key schedule shown in Fig. 3 includes a "resumption master secret" secret, which can be utilized by TLS to trigger future TLS sessions via zero round-trip resumption (0-RTT). This means that subsequent connections between client and server may benefit from enhanced triple-hybrid security without incurring in additional communication overhead.

C. Discussion and Results on Quantum-Resistant IKEv2

As the IPsec protocol heavily relies on the IKEv2 sub-protocol for the key exchange algorithm negotiation, this

TABLE III

IKEv2 KEY EXCHANGE PERFORMANCE ANALYSIS WITH DIFFERENT CRYPTOGRAPHIC (QUANTUM-RESISTANT) ALGORITHMS. RESULTS ARE PRESENTED IN MILLISECONDS (MS). DATA HAS BEEN CAPTURED ON INITIATOR'S SIDE VIA WIRESHARK [58]

IKEv2 KEX algorithm	Classical	PQ	QKD	Total
X25519	2.13	-	-	2.13
X25519-MLKEM1024	1.74	2.28	-	4.02
MLKEM1024-QKD	-	2.25	32.50	34.75
X25519-MLKEM1024-QKD	1.75	2.25	29.96	33.96

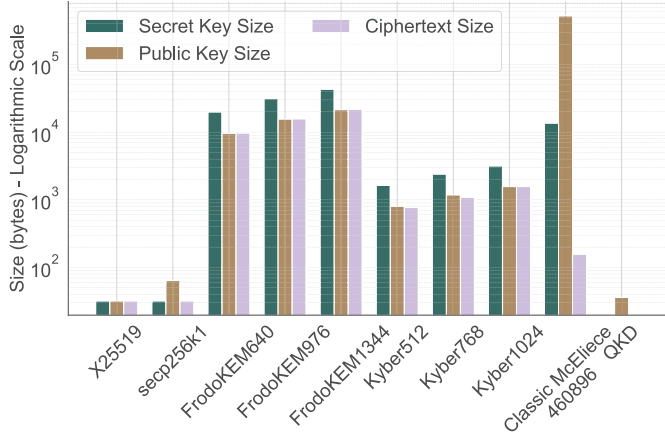


Fig. 6. Size in bytes of the different cryptographic material involved in the generation of a shared secret for the AKE [39].

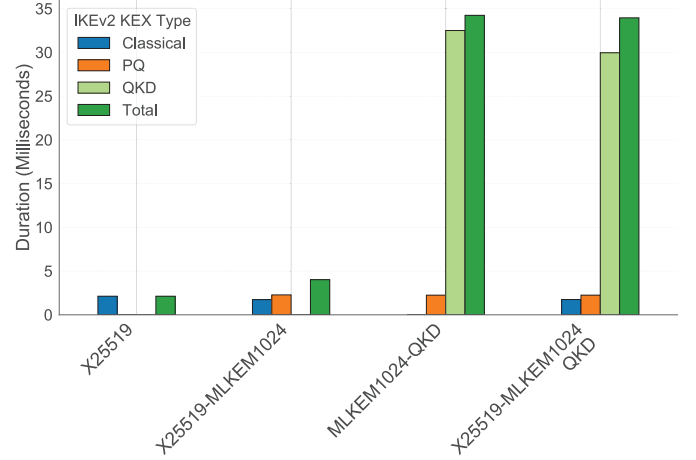


Fig. 7. Quantum-resistant IKEv2 key negotiation protocol: Key exchange performance analysis in milliseconds.

section discuss and analyses the results of the IKEv2 key negotiation under different combinations of classical, hybrid and triple-hybrid key negotiation protocols. As it happens with TLS, while the communication cost is a very important aspect in network communications, and thus new strategies to reduce communication cost for key agreement should be explored, the main motivation of this work is to analyze the impact of adding triple-hybrid encrypted communications into the IKEv2 protocol, and what is the performance-security trade-off imposed by such solutions. For benchmarking our IPsec implementation, we have employed CRYSTALS-Dilithium for the authentication, and different combinations of key exchange proposals involving the well known classical algorithm X25519, the PQ variant Kyber1204, also known as ML-KEM-1024, and our own custom implementation of QKD. In this work, the key exchange protocol X25519 is used as the control algorithm. Furthermore, as shown in Fig. 7, we provide results for a hybrid version involving classical and PQ cryptography, a fully quantum-resistant hybrid version involving PQ cryptography and QKD, and finally, a triple-hybrid IKEv2 protocol involving classical, PQ cryptography and QKD.

As it can be seen in table III, there are few differences in terms of total time needed for the key exchange negotiation. Both the single classical AKE agreement and the hybrid variant based on classical and PQ cryptography have very similar performances, mainly due to the efficiency of PQ operations involved in ML-KEM-1024, namely ephemeral PQ keypair generation, PQ key encapsulation and PQ key decapsulation. In this regard, the performance overhead is approximately

3 ms, which in the current environment can be considered negligible. Nonetheless, due to the large sizes of the PQ keys, the key exchange can no longer happen with a single packet exchange, requiring at least two more extra exchanges among initiator and responder to agree on a PQ-based ML-KEM-1024 shared secret.

Furthermore, we present a triple-hybrid quantum-resistant IKEv2 protocol enabled by classical cryptography (e.g., X25519), PQ cryptography (e.g., ML-KEM-1024) and QKD. This option aims to enhance the security of communications by integrating three different cryptographic assumptions into the *keyseed*, which will become the seed from which all the key material is derived. It must be noted the difference with the QKD key retrieval API introduced in TLS. This is due to our latest, more efficient key delivery API implemented in our in-house KMS's.

In this triple-hybrid proposal, there are four main considerations to have in mind: 1) QKD key retrieval procedure requires of an AKE among SAE and KMS in order to deliver a quantum key safely [27]. This impacts the protocol by adding an extra overhead of approximately 15 ms for collecting the QKD shared secret at both initiator and responder endpoints, which generates an extra total overhead of approximately 30 ms when agreeing on the *keyseed*. 2) Additional bandwidth is required for performing a triple-hybrid key exchange. At least two more exchanges are required for agreeing on a ML-KEM-1024 shared secret, mainly due to the longer key sizes present in PQ cryptography. Regarding QKD, only an extra is exchange,

as the only overhead added by QKD in terms of bandwidth is the transmission of the key ID among initiator responder, with a current size of 36 extra bytes. Nonetheless, this could be reduced to up to 16 bytes in future implementations via compression. 3) Currently, there is a layer of isolation among different key exchanges for each cryptographic primitive. Because of the low bandwidth requirements of QKD-based IKEv2 implementations, the QKD key agreement could be integrated together with the classical key exchange, reducing at least the overall IKEv2 agreement protocol by at least one exchange, thus reducing complexity. 4) With the presented implementation of a triple-hybrid security IKEv2 protocol, we highly enhance the security of the key agreement, and thus the whole protocol, ensuring the confidentiality of data in every possible scenario involving classical and quantum attacks, independently of their computing capacity. 5) While the time taken by the IKEv2 protocol to agree on the shared secrets augments, it must be mentioned that this procedure happens at the IPsec control plane, while the data plane where the encrypted end-user traffic is still encrypting data without interruption.

As a result, the proposed triple-hybrid key agreement impacts the performance of communications on the control plane, but its impact on the data plane performance is negligible. Thus, end-users will most likely not notice the introduction of a triple-hybrid encryption in their communications.

Finally, we present a fully quantum-resistant IKEv2 protocol based on ML-KEM-1024 and QKD. This implementation potentially gains some performance over the triple-hybrid protocol shown before, as it does not require the use of classical cryptography. Once PQ algorithms become subject to a much deeper level of research, and their security analyses become strongly accepted in the community, such hybrid quantum-resistant algorithms represent the future of encrypted communications. In this protocol, we can see a minimum gain of approximately 1 ms over the triple-hybrid version. Moreover, one less exchange of packets is required to happen among initiator and responder. For very demanding communication systems, less complexity for the key agreement and a lower number of exchanges is also a factor to consider.

D. Discussion on Quadruple and Quintuple Hybrid Key Exchanges

Following the latest PQ standards released by NIST, as well as other recommended sources for migrating towards quantum-resistant cryptography, we implement the preferred KEM algorithm – ML-KEM-1024 – as the PQ key exchange algorithm [18], [38], as well as QKD equipment from IDQuantique as the source for QKD keys. In our work, we identify that the security vs. performance trade-off of triple-hybrid key exchanges allows for such solutions to be employed in current communications.

Regarding the use of quadruple or higher key exchanges, other non-standardized PQ-based algorithms rely on the

same cryptographic assumptions as ML-KEM. Thus, although implementing quadruple or higher key exchanges via other PQ-based algorithms is technically possible with our code-base (e.g., OpenSSL, strongSwan), the overall security of the system would still rely on the same three sets of cryptographic assumptions. Therefore, we consider the diminished returns to be insufficient to justify the added complexity.

E. Discussion on Concatenation Vs. Cascade Approaches for Combining Hybrid Keys

As seen in [59], there are two primary approaches for combining key material from different sources into a single-hybrid shared secret. Concatenation and cascade.

In the concatenation-based approach, such as the TLS key schedule process shown in Section IV-B, the resulting key is a simple aggregation of the input keys. In this approach, the security of the final key material depends on the security assumptions of the different key blocks used as IKM, as well as the secure implementation of the cryptographic key derivation functions used to combine the key material (e.g., HKDF). Specifically, if we assume that the input key materials are cryptographically secure, the system's security primarily relies on the cryptographic robustness of the key derivation function. Since the concatenation approach combines all cryptographic key material before key derivation, we can assume that the system maintains its security unless all of the underlying cryptographic assumptions are compromised.

In the cascade approach, similar to the IKEv2 key derivation procedure described in V-C, the key material is iteratively updated every round until the final key material is obtained. In our triple-hybrid approach, for example, three rounds must take place before the final shared secret is obtained. In this case, a key derivation function (e.g., PRF, PRF+, HKDF) is used to combine the previously existing key material, the newly generated key material, and a seed that usually contains contextual information.

Although using the cascade approach might improve the security of the final key material, it also increases the complexity of the system, possibly generating an additional overhead as a consequence of using more cryptographic functions. Therefore, a comprehensive security vs. performance trade-off analysis must be performed.

Nonetheless, it should be noted that the key derivation procedures present in both TLS and IKEv2 are standardized. Thus, we assume that both the concatenation and cascade approaches are valid for combining key material from different sources. To comply with current standards, our TLS implementation combines QKD and PQ key material with classical cryptography via concatenation [41]. In our IKEv2 implementation, we use both concatenation and cascade approaches, as RFCs 9372 and 7296 suggest [46], [49].

VII. CONCLUSION

In an era where quantum computing threatens current cryptographic standards, our research presents a novel solution for integrating classical, PQ cryptography and QKD into two of today's most well known network security protocols: TLS 1.3 and IPsec. By combining PQ cryptography and QKD alongside classical cryptography, we achieve enhanced security against all known classical and quantum attackers, ensuring the highest level of protection for modern networks.

In TLS, our hybrid security scheme achieves quantum-resiliency at an added communication cost of approximately 57 ms during the handshake, mainly due to the performance of the key retrieval API. Moreover, our IPsec solution achieves enhanced security with minimal impact on communication, as the key negotiation agreement is carried out via the IKEv2 protocol at the control plane, while the data plane continues to encrypt data independently. Thus, the end-user benefits of an added quantum-resilience primitive with negligible added communication cost.

Our scheme supports single, hybrid or triple-hybrid security via any of the cryptographic techniques mentioned above, achieving a high degree of crypto-agility, flexibility and quantum-resiliency very much needed in the upcoming quantum era. The crypto-agile nature of our solution, as well as the integration with widely-known open source libraries (e.g., strongSwan, OpenSSL) for encryption at layers 3 and 4 addresses some of the biggest challenges of migrating to quantum-resistant communications: it reduces potential vulnerabilities of PQ cryptography, allows a quick and agile replacement of any of the components involved and guarantees long-term confidentiality by implementing future-proof cryptographic schemes.

Our solution protects against quantum-based “harvest now, decrypt later attacks”, removing any concerns regarding the maturity and security of PQ cryptography or QKD. By proactively addressing the quantum threat by combining classical, PQ cryptography and QKD in a seamless way, our work helps future-proof communication systems, providing a bridge between current security protocols and the post-quantum era, and paving the way towards the standardization of triple-hybrid quantum-resistant communication systems.

VIII. FUTURE WORKS

Regarding future works, several challenges need to be overcome in order to enable the widespread adoption of the presented solution in communication networks. First, the throughput requirements of current networks are increasing exponentially in recent years, mainly due to the large amounts of data being exchanged over the Internet. A significant challenge that is not addressed in this paper is meeting these throughput requirements at the data plane level. To address this, the presented quantum-resistant solutions, both TLS 1.3 and IPsec, must be integrated with high-speed data encryption techniques, such as those demonstrated in [60].

Moreover, further developments in the integration of classical and QKD networks are essential to enable the coexistence of classical and quantum signals in the same physical medium

without significantly reducing the SKR. This advancement would allow the presented solution to be deployed within existing telecommunications infrastructure, eliminating the need for deploying costly dedicated dark networks solely for QKD. In this context, the development of software-defined QKD networks promises to facilitate and orchestrate the integration of classical and quantum networks over the same physical medium [23].

Finally, due to the physical requirements of QKD, our triple-hybrid solution is only applicable in network segments where quantum links — either optical or free-space based — are available. However, as proposed in [61], alternative approaches must be employed for network segments where quantum links are not feasible. These alternative solutions will likely employ a combination of classical and PQ cryptography. Further research is necessary to seamlessly integrate QKD-based and non-QKD-based solutions, ensuring that end-to-end communications are fully protected via quantum-resistant communication protocols.

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